

Original Mixed-radix: What Worked and What Did Not

A real mechanism, but no useful natural-data frontier improvement

2026-08-01

1. The Answer First

Final result for the tested method

The original fixed-adjacent mixed-radix method is **not competitive on SIFT1M or GIST1M** under the frozen Recall–speed evaluation.

- At 32 bytes, it becomes exactly the power-of-two control.
- At 64 bytes, Recall changes by at most about one tenth of one percentage point, with both positive and negative signs.
- A synthetic positive control now confirms that the intended mechanism itself is real and reaches the query result.

Interpretation

The implementation can exploit non-power-of-two cardinalities. The tested natural data simply does not contain enough of this structure for a useful win.

2. Scope Correction: Which “Attempt 4” Is This?

Original target: this deck

Two adjacent scalar quantizers share one packed word. Their numbers of levels may be arbitrary positive integers.

Separate branch: not this result

The structured-2D method learns a reusable two-dimensional shape with affine transforms. It was evaluated separately and is currently set aside.

Why the distinction matters

Mixed-radix changes how a fixed number of bit patterns is divided between two scalar coordinates. It does **not** learn a shared 2D vector codebook.

3. The Original Intuition

Four bits give exactly $2^4 = 16$ possible labels for one coordinate pair.

Power-of-two allocation

Each coordinate receives 1, 2, 4, 8, ... levels. A typical balanced split is $4 \times 4 = 16$.

Mixed-radix allocation

Allow any integers K_1, K_2 as long as

$$K_1 K_2 \leq 16.$$

This includes $3 \times 5 = 15$.

Hypothesis

If one coordinate really needs 3 values and the other needs 5, mixed-radix can retain all 15 combinations. Separate binary rounding would need $4 \times 8 = 32$ states and cannot fit in four bits.

4. Running Example: Pack a 3×5 Grid

Let $z_1 \in \{0, 1, 2\}$ and $z_2 \in \{0, 1, 2, 3, 4\}$. Store

$$\text{label} = z_1 + 3z_2.$$

	$z_2 = 0$	1	2	3	4
$z_1 = 0$	0	3	6	9	12
$z_1 = 1$	1	4	7	10	13
$z_1 = 2$	2	5	8	11	14

Every real combination has a unique label; label 15 is simply unused.

What the power-of-two control must do

With only 16 labels it may choose 4×4 , but then five distinct values of the second coordinate must share only four reconstructed values. At least two real prototypes can receive the same stored code.

5. Training, Encoding, and Querying

Training, using base/index data only:

- for each adjacent coordinate pair:

 - estimate scalar reconstruction cost for each level count

 - choose K_1 , K_2 minimizing total cost, subject to $K_1 * K_2 \leq 2^B$

Encoding one database vector:

- z_1 = nearest scalar centre on coordinate 1

- z_2 = nearest scalar centre on coordinate 2

- store label = $z_1 + K_1 * z_2$

Querying:

- build the complete 2^B -entry distance table for each pair

- decode one label per pair, add table values, keep the best 100 IDs

Controlled difference

Arbitrary arm A and power-of-two arm D use the same pairs, bytes, candidates, table builder, and scan loop. Only the allowed values of K_1 , K_2 differ.

6. What We Actually Compared

A128

D128 FULL

PQ / OPQ

data

payload

index sizes

quality

speed

search range

arbitrary integer radices, such as $15 \times 17 = 255$

identical method, but each radix must be a power of two

standard product-quantization baselines for deployability context

SIFT1M and GIST1M

exactly 32 or 64 code bytes per database vector

1,024 and 4,096 coarse database cells

Recall@100: how many of the true nearest 100 are returned

single-query latency and 12-core queries per second

nine fixed numbers of cells searched per query

Fairness rule

A and D search exactly the same candidates and perform the same table work. The allocation was fixed before reading query results.

7. What the Natural-data Allocator Chose

32-byte vectors: 4 bits per pair

Every A group selected 4×4 . Therefore A and D were literally the same representation and returned identical rankings.

64-byte vectors: 8 bits per pair

Only 10–24 of 64 groups differed from 16×16 . Almost all were 15×17 or 17×15 , using 255 of 256 paid states.

Early mechanism diagnosis

The reconstruction-based allocator found no large unused-cardinality problem on SIFT/GIST. At 64 bytes it mostly moved only one level from one coordinate to its partner.

8. Natural-data Recall Result

At the same number of searched cells, A-minus-D Recall@100 was:

Dataset	coarse cells	minimum	maximum
GIST1M	1,024	-0.00096	+0.00054
GIST1M	4,096	-0.00106	+0.00017
SIFT1M	1,024	-0.000097	+0.000066
SIFT1M	4,096	-0.000033	+0.000436

Meaning in plain language

The largest gain was 0.054 percentage points; the largest loss was 0.106 percentage points. The sign changed across datasets, index sizes, and search ranges, so there is no stable quality improvement.

9. Speed and Baseline Result

- A/D throughput ratio: 0.9918 to 1.0069; group medians are essentially 1.
- A/D p95 latency ratio: 0.9897 to 1.0251.
- This is expected: A and D deliberately execute the same complete-table consumer and nearly the same representation.
- PQ and OPQ generally reach higher Recall in the useful high-quality region.

Natural-data decision

The fixed-adjacent mixed-radix method did not move a useful Recall–speed frontier. A tiny nonzero difference is not a database-systems contribution.

Complete matrix: 512 accepted passes; 32.36 CPU-hours; 17.09 wall-hours; all eight correctness tests pass.

10. Why Add a Synthetic Positive Control?

The natural result left two different explanations:

- 1 the mixed-radix mechanism is valid, but the required data structure is absent or weak;
or
- 2 the implementation cannot transmit a radix advantage into query ranking, even when such an advantage exists.

Decisive construction

Create exact Cartesian prototypes with the cardinalities required by the intuition. Put 100 identical database vectors at every prototype, place one query exactly on each prototype, search every candidate, and ask whether all 100 true duplicates are recovered.

Positive cases are 3×5 in 4 bits and 15×17 in 8 bits. Matching 4×4 and 16×16 cases are zero-effect controls.

11. Synthetic Result: The Mechanism Reaches Recall

Case	A shape	D shape	code collisions A/D	Recall A/D
positive B4	3×5	4×4	0 / 3	1.000 / 0.800
null B4	4×4	4×4	0 / 0	1.000 / 1.000
positive B8	15×17	16×16	0 / 15	1.000 / 0.941
null B8	16×16	16×16	0 / 0	1.000 / 1.000

Validity checks

In both null controls A and D produced identical rankings. In both positive cases the rankings differed exactly where D merged prototypes. Two complete runs were byte-for-byte identical.

12. Put the Two Results Together

Question	Natural SIFT/GIST	Synthetic witness
Can A choose non-dyadic radices?	rarely, and only weakly at 64 bytes	yes, exactly 3×5 and 15×17
Does this avoid collisions?	no material measured benefit	yes, 3 and 15 control collisions removed
Does it improve Recall?	tiny, mixed-sign changes	yes, by 0.200 and 0.0588
Does it beat PQ/OPQ?	no useful frontier movement	not tested or claimed

Combined conclusion

The mechanism is real but its necessary structure is not sufficiently strong in the tested natural representation. The synthetic result explains the negative result; it does not reverse it.

13. Claim Boundary and Next Question

What we can now say

Fixed-word mixed-radix coding can preserve cardinalities that binary rounding cannot, and this can prevent query-result collisions.

What we cannot say

We have not shown that the condition is common in real vector data, that it improves end-to-end QPS, or that it beats an unrestricted PQ/VQ model.

If we continue

The next distinct question is whether base data alone can identify coordinate groups whose quantization collisions affect near-neighbour ordering. First compute an oracle upper bound; do not immediately build another large system.

References and Evidence

- Jégou, Douze, and Schmid. "Product Quantization for Nearest Neighbor Search." TPAMI 2011.
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- Gao and Long. "RaBitQ." SIGMOD 2024.
- Li et al. "SAQ." 2025.
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- Synthetic mechanism result: `mixed_radix_synthetic_witness_result_2026_08_01.md`.